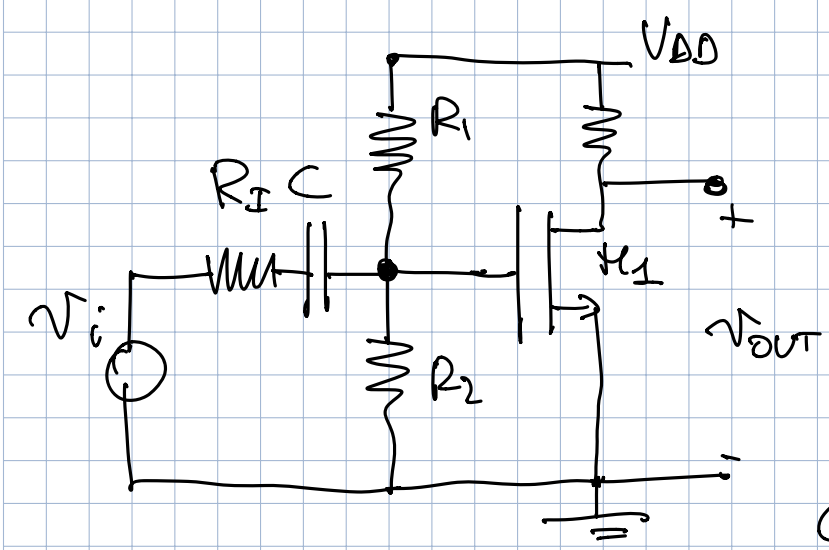
ESERCIZIO PROPOSTO



$V_{DD} = 12V$

$R_1 = 270\ K\Omega$ $\quad R_2 = 220\ K\Omega$

$R_D = 4{,}7\ K\Omega$ $\quad R_i = 10\ K\Omega$

$V_{TN} = 3V$ $\quad K_n = 0{,}5\ mA/V^2$

$\lambda = 0{,}01\ V^{-1}$

$v_i = V_s \cos \omega t$

$V_s = 10\ mV$

CALCOLARE DC, AC, A_v, v_{OUT} E VERIFICARE SE PICCOLO SEGNALE È RISPETTATO

[$V_G = 5{,}39V$, $I_{DQ} = 1{,}43\ mA$, $V_{DSQ} = 5{,}3V$, $g_m = 1{,}2\ mS$

$r_o = 73{,}9\ K\Omega$, $A_v = -4{,}87$, $v_{OUT} = 5.3V - 48{,}7 \cos \omega t\ [mV]$]

$v_{gs} < 0{,}2\,(V_{GS} - V_{TN}) = 0{,}2\,(2{,}39V) = 0{,}478V$; $v_{gs} = 9{,}24\ mV$ OK